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B.Tech. CSE Trimester -5

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Generative AI
25CS2207E

LAB RECORD

2026

WEEK 1 – FIRST LLM API CALL USING OPENAI

1. Aim

To establish a connection between a Python application and the OpenAI platform by configuring the API key and installing the necessary Python libraries.

2. Requirements

- Python 3.14
- PyCharm IDE
- OpenAI Platform account
- OpenAI API key
- OpenAI Python SDK
- python-dotenv library
- Internet connection

3. Theory

OpenAI provides an API that allows developers to use its language models in their applications. Through this interface, a Python program can submit text inputs and receive AI-generated outputs.

An API key is a confidential authentication token that grants an application access to an API. It is important to keep this key secure and avoid exposing it in the source code.

The OpenAI Python SDK provides the necessary functions to communicate with OpenAI models. The `python-dotenv` package helps load confidential information stored in a `.env` file.

4. Procedure

1. Created a new Python project in PyCharm.
2. Configured the project using a Python virtual environment.

3. Opened the OpenAI Platform and created an account.
4. Generated an API key for the project.
5. Created a `.env` file inside the project directory.
6. Added the API key to the file using the variable `OPENAI_API_KEY`.
7. Installed the required dependencies using:
   `python -m pip install openai python-dotenv`
8. Created a Python source file named `openai_call.py`.
9. Imported the OpenAI SDK, operating system module, and dotenv library.
10. Loaded the environment variables and accessed the API key using `os.getenv()`.
11. Wrote the code to initialize the OpenAI client.
12. Examined the API billing requirements. Since billing was not completed, the actual API call was not performed.

5. Code

```
import os
from dotenv import load_dotenv
from openai import OpenAI

# Load environment variables
load_dotenv()

# Get the API key
api_key = os.getenv("OPENAI_API_KEY")

if not api_key:
    raise ValueError("OpenAI API key not found!")

# Initialize the client
client = OpenAI(api_key=api_key)

# Send input to the model
response = client.responses.create(
    model="gpt-5.5",
```

```
    input="What is Artificial Intelligence? Explain in simple words."
)

# Print the model response
print("\nLLM Response:")
print(response.output_text)
```

Note: The code was prepared, but execution was postponed because API billing was not configured.

6. Observation

- The OpenAI API key was generated successfully.
- The required Python packages were installed in the virtual environment.
- The `.env` file was configured to store the authentication key.
- The Python script was prepared to initialize the OpenAI client.
- It was understood that API access and ChatGPT access are different services.
- API requests may require an active billing setup or available credits.

7. Result

The initial setup for using the OpenAI API with Python was completed. The API key configuration, library installation, and client initialization code were prepared successfully. The actual model response generation is pending due to incomplete billing configuration.

